

# Core Pure Mathematics 1

Notes · Edexcel A Level Further Mathematics — Core Pure 1 · Chapter 3: Series

## Chapter 3: Series

### Why this matters

You already know how to sum arithmetic and geometric series from A Level Maths. This chapter introduces genuinely new machinery: standard formulae for summing powers of  $r$  ( $r$ ,  $r^2$ ,  $r^3$ ), and the method of differences for series that don't fit any standard pattern at all. Both techniques reappear constantly in Chapter 8 (Proof by Induction), so fluency here pays off twice.

### Concept: standard summation formulae

Three results, all provable by induction (see Chapter 8), let us sum the first  $n$  natural numbers, their squares, and their cubes in closed form:

$$\begin{aligned}\sum_{r=1}^n r &= n(n+1)/2 \\ \sum_{r=1}^n r^2 &= n(n+1)(2n+1)/6 \\ \sum_{r=1}^n r^3 &= [n(n+1)/2]^2 = (\sum r)^2\end{aligned}$$

These are given in the formula booklet, so **the skill being tested is manipulation, not memorisation**: splitting more complicated sums into combinations of these three using linearity ( $\sum(ar + b) = a\sum r + \sum b$ ), then factorising the result fully.

#### Key vocabulary

| Term                      | Meaning                                                                     |
|---------------------------|-----------------------------------------------------------------------------|
| $\Sigma$ (sigma) notation | Shorthand for a sum: $\sum_{r=1}^n f(r)$ means $f(1) + f(2) + \dots + f(n)$ |
| Closed form               | An expression for a sum in terms of $n$ , with no "..."                     |
| Method of differences     | Rewriting each term as a difference so that a sum telescopes                |
| Telescoping               | Consecutive terms cancelling in pairs, leaving only a few surviving terms   |

#### Worked example 1: using the standard formulae directly

##### EXAMPLE 1

**Question.** Find  $\sum_{r=1}^n r(r+1)$ , giving your answer as a fully factorised expression in  $n$ .

**Step 1.** Expand the summand:  $r(r+1) = r^2 + r$ . **(M1)**

**Step 2.** Split the sum using linearity:  $\sum(r^2 + r) = \sum r^2 + \sum r$ . Then substitute the standard formulae for  $\sum r^2$  and  $\sum r$ :  $\sum r^2 + \sum r = n(n+1)(2n+1)/6 + n(n+1)/2$ . **(M1 A1)**

**Step 3.** Take out the common factor  $n(n+1)/6$ :  $n(n+1)(2n+1)/6 + n(n+1)/2 = [n(n+1)/6] \times [(2n+1) +$

3]. **(M1 A1)**

**Step 4.** Simplify:  $= [n(n+1)(2n+4)]/6 = n(n+1)(n+2)/3$ . **(A1)**

**Answer:**  $\sum_{r=1}^n r(r+1) = n(n+1)(n+2)/3$ .

#### Exam tip

##### EXAM TIP

Always find the common factor (usually  $n(n+1)$  or  $n(n+1)/6$ ) BEFORE attempting to simplify the bracket. Trying to expand everything into a single messy polynomial and then re-factorise from scratch is where most marks are lost — keeping  $n(n+1)$  visible throughout makes the final factorisation almost automatic.

*Quick check:* testing any factorised summation formula at a small value of  $n$  (e.g.  $n = 1$  or  $n = 2$ ) against the sum computed directly is a fast way to catch an algebra slip before moving on.

#### Worked example 2: sums starting from a value other than 1, or in two variables

##### EXAMPLE 2

**Question.** Find  $\sum_{r=5}^{10} r^2$  in terms of  $n$  where needed, i.e. evaluate this specific numerical sum.

**Step 1.** Use the fact that  $\sum_{r=5}^{10} r^2 = \sum_{r=1}^{10} r^2 - \sum_{r=1}^4 r^2$  (sum up to 10, minus the part you don't want, up to 4). **(M1)**

**Step 2.** Apply the formula with  $n = 10$ :  $\sum_{r=1}^{10} r^2 = 10(11)(21)/6 = 385$ . **(A1)**

**Step 3.** Apply the formula with  $n = 4$ :  $\sum_{r=1}^4 r^2 = 4(5)(9)/6 = 30$ . **(A1)**

**Step 4.** Subtract:  $385 - 30 = 355$ . **(A1)**

**Answer:**  $\sum_{r=5}^{10} r^2 = 355$ .

#### Exam tip

##### EXAM TIP

This "sum up to the top, minus the sum up to one below the bottom" trick is easy to forget under exam pressure because it looks so different from the algebraic manipulation in worked example 1. Give it a memorable name (e.g. "chop the bottom off") so it becomes an instantly recognisable pattern rather than something you have to re-derive.

### Concept: method of differences

Some series cannot be summed using the standard formulae because each term is a fraction or otherwise doesn't reduce to a polynomial in  $r$ . The **method of differences** rewrites the general term  $u_r$  as  $f(r) - f(r+1)$  (or similar), so that when the sum is written out term by term, almost everything cancels ("telescopes"), leaving only a handful of boundary terms.

Picture it like a row of dominoes standing on end, each one both knocking over and being knocked over by its neighbour: every  $f(r)$  term is cancelled by an  $f(r+1)$  appearing with the opposite sign two

terms later, all the way down the sum, so only the very first and very last dominoes — which have nothing left to cancel them — are still standing at the end.

### Worked example 3: method of differences

#### EXAMPLE 3

**Question.** Given that  $1/(r(r+1)) = 1/r - 1/(r+1)$ , find  $\sum_{r=1}^n 1/(r(r+1))$  in terms of  $n$ .

**Step 1.** Use the given identity to rewrite the sum:  $\sum_{r=1}^n [1/r - 1/(r+1)]$ . **(M1)**

**Step 2.** Write out the first few and last few terms explicitly to see the pattern:  $(1/1 - 1/2) + (1/2 - 1/3) + (1/3 - 1/4) + \dots + (1/(n-1) - 1/n) + (1/n - 1/(n+1))$ . **(M1)**

**Step 3.** Observe that every term cancels with the next except the very first  $(1/1)$  and the very last  $(-1/(n+1))$ . **(A1)**

**Step 4.** Write the remaining terms:  $1 - 1/(n+1) = (n+1)/(n+1) - 1/(n+1) = n/(n+1)$ . **(M1 A1)**

**Answer:**  $\sum_{r=1}^n 1/(r(r+1)) = n/(n+1)$ .

#### Exam tip

##### EXAM TIP

Always write out at least three terms at the start AND at least two terms at the end (never just "...") before cancelling. This is the step examiners specifically look for method marks on, and it is also exactly where it is easy to accidentally leave in a term that should have cancelled, or cancel a term that shouldn't.

### Worked example 4: a trickier method-of-differences identity

#### EXAMPLE 4

**Question.** Show that  $r^3 - (r-1)^3 = 3r^2 - 3r + 1$ , and use this result to find  $\sum_{r=1}^n r^2$  by the method of differences.

**Step 1.** Expand  $(r-1)^3 = r^3 - 3r^2 + 3r - 1$ , so  $r^3 - (r-1)^3 = 3r^2 - 3r + 1$ , as required. **(M1 A1)**

**Step 2.** Sum both sides from  $r = 1$  to  $n$ . The left-hand side telescopes:  $\sum_{r=1}^n [r^3 - (r-1)^3] = n^3 - 0^3 = n^3$ . **(M1 A1)**

**Step 3.** The right-hand side is  $3\sum r^2 - 3\sum r + \sum(1) = 3\sum r^2 - 3 \times n(n+1)/2 + n$ . **(M1)**

**Step 4.** Set the two sides equal and solve for  $\sum r^2$ :  $n^3 = 3\sum r^2 - 3n(n+1)/2 + n$ , rearranging to  $\sum r^2 = [n^3 - n + 3n(n+1)/2] / 3$ , which simplifies to  $n(n+1)(2n+1)/6$  after collecting over a common denominator. **(M1 A1)**

**Answer:**  $\sum_{r=1}^n r^2 = n(n+1)(2n+1)/6$ , confirming the standard formula — this is exactly how the formula is derived in the first place.

## Chapter summary

| Idea | Key point |
|------|-----------|
|------|-----------|

|                              |                                                                                                     |
|------------------------------|-----------------------------------------------------------------------------------------------------|
| <b>Standard formulae</b>     | $\Sigma r$ , $\Sigma r^2$ , $\Sigma r^3$ are given in the formula book; use linearity to split sums |
| <b>Common factor first</b>   | Factor out $n(n+1)/6$ (or similar) before simplifying the remaining bracket                         |
| <b>Sums from a to b</b>      | $\Sigma_{r=a}^b = \Sigma_{r=1}^b - \Sigma_{r=1}^{a-1}$                                              |
| <b>Method of differences</b> | Rewrite $u_r$ as $f(r) - f(r+1)$ ; write out terms explicitly and cancel                            |
| <b>Telescoping</b>           | Only boundary terms survive; always show enough terms to justify this                               |

#### Common exam mistakes

- Forgetting to factor out  $n(n+1)$  early, leading to messy, unfactorisable expressions.
- Getting the limits wrong when adjusting a sum that doesn't start at  $r = 1$ .
- Writing only "..." in the method of differences without showing enough individual terms to justify which ones cancel.
- Sign errors when the difference is  $f(r+1) - f(r)$  rather than  $f(r) - f(r+1)$  (the direction affects which terms survive).
- Not fully simplifying the final answer into a single fraction or fully factorised form.

#### You should now be able to

- Use and manipulate the standard formulae for  $\Sigma r$ ,  $\Sigma r^2$ , and  $\Sigma r^3$ .
- Evaluate sums that do not start at  $r = 1$  by subtracting an unwanted lower portion.
- Apply the method of differences to sum a series given a suitable identity.
- Derive one standard summation formula from another using the method of differences.